Year 10 General Science

Energy and Change Investigation – Levers

NAME: _____ **Mark** _____ **/ 38**

OTHER GROUP MEMBERS: _____

_____

Aim

To investigate how changing the effort affects the effort arm when lifting a load.

Hypothesis What do you think will happen? Explain why.
(2 Marks)

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Equipment

Meter rule, wedge to act as a fulcrum, plasticine, brass masses

Method

Which variable(s) are you going to:

- Change? (independent)
  (1 Mark)

_____

- Measure? (dependent)
  (1 Mark)

_____

- Keep the same? (controlled)
  (2 Mars)

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- How will you make it a fair test?
  (3 Marks)

- List the steps you will take to do the investigation and collect the results.

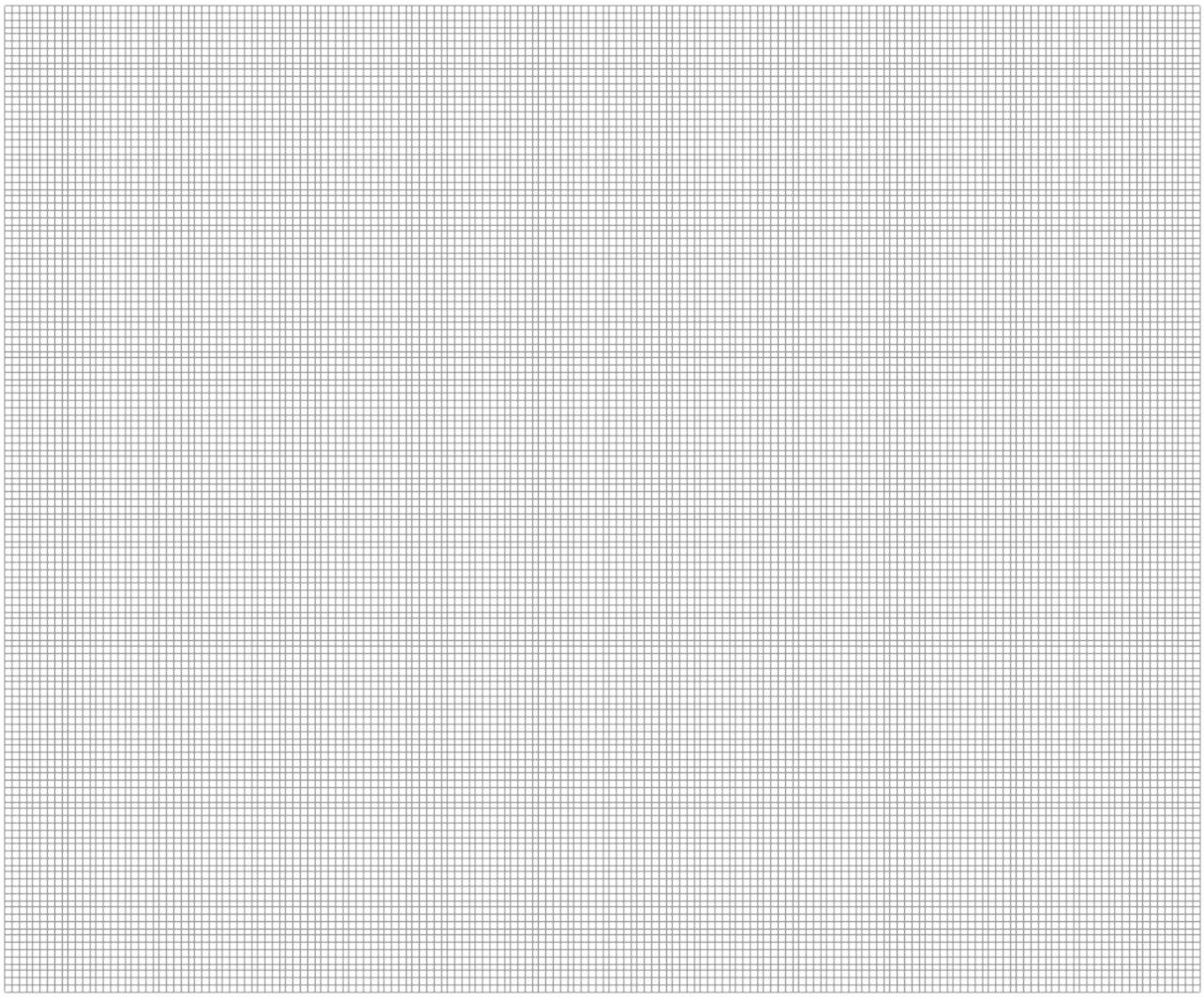
1. *Set up the metre rule so it is balancing on the wedge at the 50 cm mark. You may need plasticine to ensure that it is well balanced.*
2. *Place five 50g brass masses at the 40 cm mark. This is the* ***load.***
3. *Place four brass masses on the other side of the 50 cm mark. This is the* ***effort****.*
4. *Slowly move these masses away from the centre until the rule balances.*
5. ***Record*** *the distance in cm from the* ***centre*** *of the rule* ***to these four masses****.*
6. ***Repeat*** *step 3 with 3, 2 and then one brass mass. Record in the table below.*

Results

Record your results or observations. Use a table if appropriate.
(4 Marks)

Number of brass masses **(effort)**	Total mass of effort (g)	Distance from the centre of rule to effort **(effort arm)** (cm)
4		
3		
2		
1		

Can your results be represented in a graph?
(5 Marks)



Discussion

What do your results tell you? Are their any patterns or trends in your results?
(2 Marks)

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Critical Analysis

Using some *Science ideas*, can you explain *why* you got these patterns or trends?
(4 Marks)

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What things *went wrong* or *affected the results* you obtained? Suggest *improvements* for this investigation if possible.
(3 Marks)

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Conclusion

- What is the *relationship* you found from this investigation? It should *answer the question asked in the "aim".*
  (5 Marks)

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Questions

- Describe the shape of the trend or pattern suggested by your graph.

  (2 Marks)

  _____

  _____

- Why were the load and load arm fixed in this investigation?

  (2 Marks)

  _____

  _____

- What type of lever did you use in this investigation? Explain.

  (2 Marks)

  _____

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